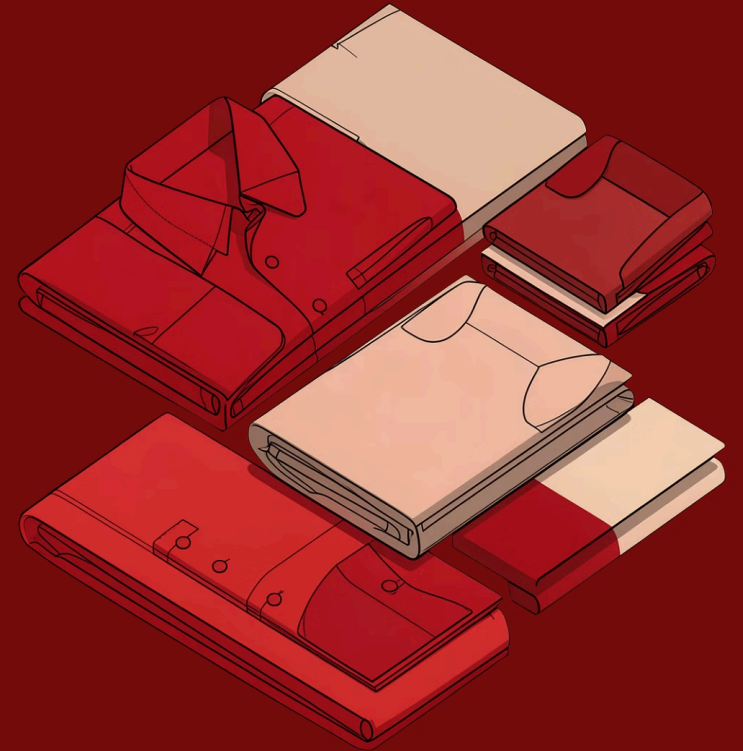


Fashion-MNIST Image Classification Using CNN

A deep learning approach to automated fashion product classification using Convolutional Neural Networks.



Project Overview

Dataset

Fashion-MNIST — 28×28 grayscale images across **10 clothing categories**

Objective

Build a CNN to **automatically classify** fashion products from raw pixel data

Framework

TensorFlow/Keras with **Adam optimizer** for gradient-based learning



Category
#000000

Category
#000000

Category
#000000

Category
#000000



Category
#000000



Category
#000000



Category
#000000



Category
#000000



Category
#000000



Category
#000000



Category
#000000



Category
#000000



Category
#000000



Category
#000000



Category
#000000



Category
#000000

The Problem

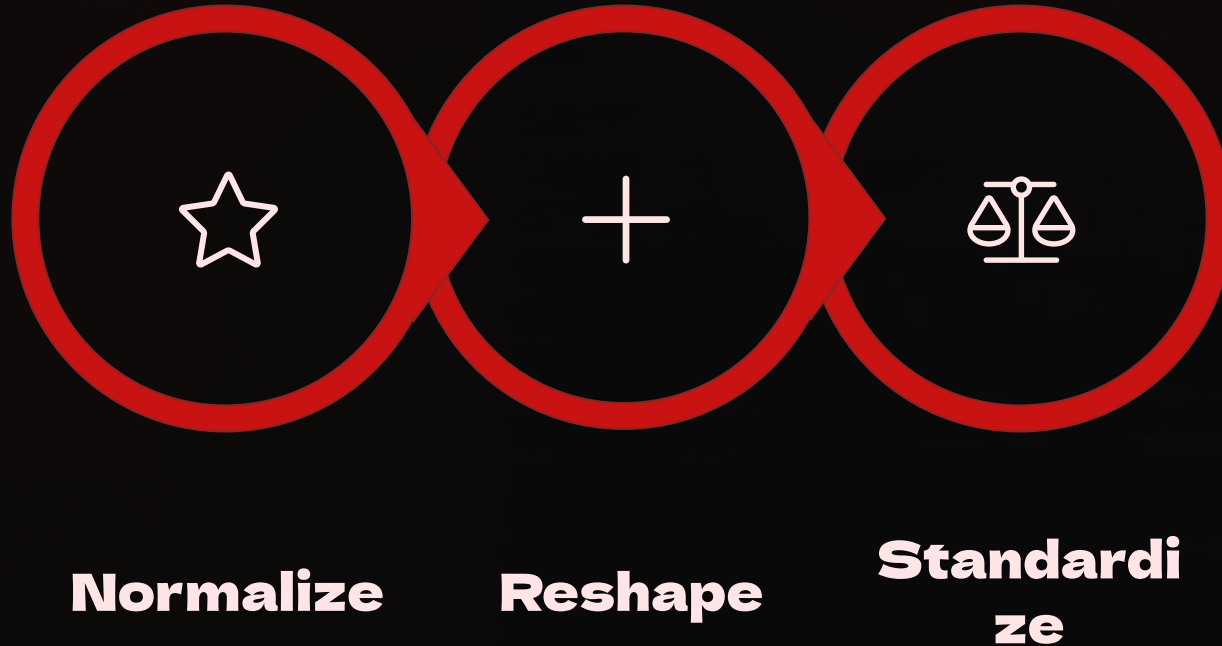
Classification Task

Map each 28×28 grayscale image to one of 10 predefined fashion categories — with no hand-crafted feature engineering.

Why CNNs?

Convolutional Neural Networks automatically learn spatial and visual features directly from raw image data, making them uniquely suited for visual classification problems.

Data Preprocessing Pipeline



Each step ensures consistent, well-conditioned inputs before any learned transformation is applied.

CNN Architecture: Core Workflow



Convolution

Learnable filters detect edges, shapes, and complex visual patterns



ReLU

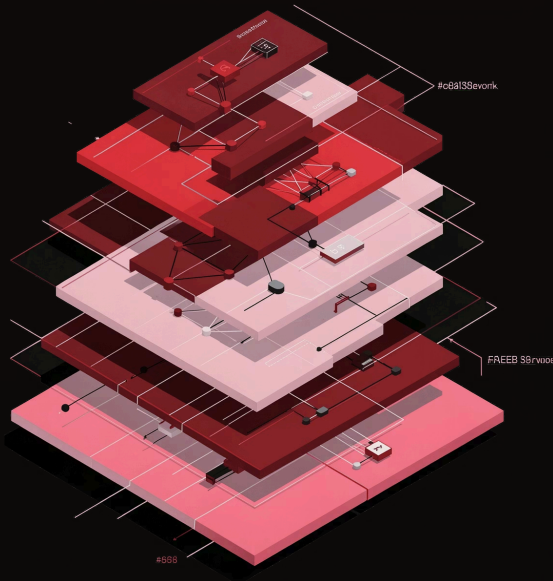
Introduces non-linearity by replacing negative values with zero



Pooling

Max pooling reduces spatial dimensions while preserving strong features

Feature Extraction & Classification



Flatten

Feature maps are converted into a one-dimensional vector, bridging convolutional and dense layers.

Dense Layers

Fully connected layers learn high-level classification patterns. Dropout is applied optionally for regularization to prevent overfitting.

Output

10 units with Softmax activation — each unit outputs a class probability, and the highest-confidence prediction determines the label.



Model Training

Adam Optimizer

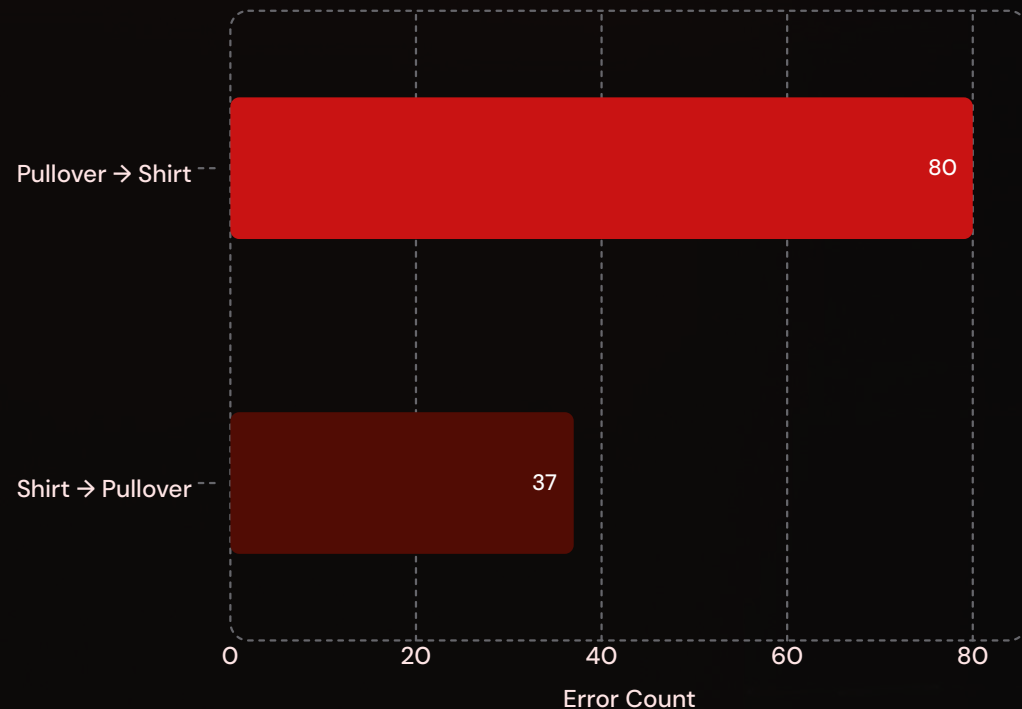
Performs adaptive gradient-based parameter updates, adjusting learning rates per weight for faster convergence.

Iterative Improvement

The model calculates classification loss each epoch and backpropagates gradients to systematically improve predictions on labeled training data.

Model Evaluation & Error Analysis

Confusion Direction



Overall Accuracy

The primary metric is the proportion of correctly classified test images across all 10 categories.

Class-Specific Error Analysis

The most significant confusion occurs between visually similar categories:

- **80** Pullover images misclassified as Shirt
- **37** Shirt images misclassified as Pullover
- **117 total** cross-category errors between these two classes

Key Limitations



Resolution Constraints

At just 28×28 pixels, the grayscale images carry limited detail. Fine-grained texture and structural cues that distinguish similar garments are largely lost at this resolution.

Inter-Class Visual Similarity

Categories such as **Pullover** and **Shirt** share similar silhouettes and textures, making the decision boundary difficult for the model to learn reliably — as reflected in the 117 observed misclassifications.

Future Improvements



Data Augmentation

Flipping, rotation, and cropping to expand effective training set diversity and reduce overfitting



Hyperparameter Tuning

Systematic search over learning rates, filter sizes, and layer depths — including batch normalization and learning-rate scheduling



Deeper Architectures

Explore residual connections and deeper stacks to capture richer hierarchical features from limited-resolution inputs



Confusion Matrix Analysis

Comprehensive per-class error breakdown to surface additional misclassification patterns beyond Pullover/Shirt